Adam Berker for Ethereum Institutional

REGULATORY SCHEMA · SCORING + REGIME DETAIL

Tokenized Bonds — Regulated Jurisdictions Full Review

The full launch matrix — eight axes across ten jurisdictions, each cell with its basis — followed by the detailed regime brief for every jurisdiction: what the issuer must do, the documents and infrastructure it needs, the network requirements, cost and subsidies, and the deals that prove it. A closing section shows how each requirement is met when a bond is launched on public Ethereum.

AUTHOR

Adam Berker

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RATING

- STRONG — favourable — low friction, or proven
- PARTIAL — workable, with trade-offs
- LIMITED — heavy, or early-stage
- NONE — absent

RECORD-KEEPER MODEL

- A — entry is the record — no interposed licensee
- B — a licensed record-keeper is interposed
- C — a regulated depository holds the record

Friction axes (clarity, ease, cost) run favourable → heavy; capability axes (public chain, precedents, cash leg, incentives) run present → absent. Each matrix cell and each regime brief states its basis.

PART ONE

I The scoring matrix.

Eight axes across ten jurisdictions. Each cell states the basis for its rating. Read the friction axes first — clarity, ease, cost and time — then the capability axes that decide whether a deal can actually be done.

The decision shortcut.

Which one wins depends on a single branch — whether you need a benchmark bearer bond and/or a listing with a secondary market. For an unlisted private placement, the ranking that follows holds; change the requirement and the pick changes.

DEFAULT · UNLISTED, EU, PUBLIC CHAIN

France

Clear, light, cheap; issuer keeps the register on public Ethereum.

LEANEST · EU ACCESS NOT REQUIRED

Switzerland

A registration agreement is the whole mechanism; strongest cash leg (SNB wCBDC).

TURNKEY + LISTING · PAY FOR SPEED

Luxembourg

Ready ecosystem and a LuxSE route; a regulated record-keeper is mandatory.

LOWEST CASH COST + STRONG CASH LEG

Hong Kong

Subsidised issuance and tokenised-CBDC settlement — on a permissioned rail.

NEED BEARER / BENCHMARK NATIVELY

Germany

Bearer-native and listable in the base regime — heavier to run.

READING THE BRANCH

A benchmark bearer bond or a listing is not prohibited in France or Switzerland — it simply leaves the lean base path. France routes it through the EU DLT Pilot Regime (since 2023); Switzerland through SDX. Germany is the only regime here that does bearer and listing in its base framework. Across the EU, secondary trading and settlement at scale run through the DLT Pilot Regime.

Friction axes.

How hard the regime is to use: whether the law is settled, how many licensed parties must be interposed, and what it costs in money and time to launch and run.

JURISDICTION & LEGAL BASIS	CLARITY law settled / bespoke	EASE parties, licences	COST & TIME launch & run
01 France [A] DEEP · Ord. 2017-1674 · Décret 2018-1226	STRONG Settled 2017; ratified; book-entry equivalence	STRONG Issuer is registrar by default; no licensee	STRONG No registrar fee; private placement avoids prospectus
02 Switzerland [A] DLT Act · CO Art. 973d (2021)	STRONG Settled 2021; ledger-based securities defined	STRONG A registration agreement is the whole mechanism	STRONG Leanest for unlisted; SDX only if listed
03 Luxembourg [B] Blockchain I–IV · Law of 2013, am. 2024	STRONG Mature EU framework; deep ecosystem	PARTIAL Mandatory bank/firm record-keeper (existing licence)	PARTIAL Fast to market; premium service fees
04 Hong Kong [C] Existing HK law + DBGS grant	PARTIAL Existing law + gov lead; not bespoke certainty	PARTIAL CMU + platform + syndicate; gov-oriented	STRONG DBGS subsidy offsets issuance cost
05 Germany [B] eWpG §16 · registrar under KWG (2021)	STRONG Bespoke crypto-securities law; tested	LIMITED Mandatory licensed registrar + separate custodian	LIMITED Most parties to engage; higher coordination
06 United Kingdom [C] Digital Securities Sandbox · BoE + FCA (2024)	PARTIAL Temporary sandbox, not permanent law	PARTIAL Approved depository; limited live operators	PARTIAL Sandbox admission; ecosystem maturing
07 Singapore [C] Tech-neutral SFA · MAS Guardian / GL1	PARTIAL Tech-neutral SFA; register rests on existing law	PARTIAL Standard roles; commercial path (GL1) forming	PARTIAL Pilot-stage; no turnkey route yet
08 UAE [C] SCA (onshore) · DFSA / FSRA (free zones)	PARTIAL Frameworks in force; thin bond precedent	PARTIAL ADX + HSBC Orion / CMU	PARTIAL Business-friendly; bond path is new
09 United States [B] Existing securities law · SEC transfer-agent rules	PARTIAL No bespoke law; clear for funds, thin for bonds	LIMITED Transfer agent + custodian + broker-dealer + trustee	LIMITED Heavy structuring; expensive counsel
— EU-wide [layer] DLT Pilot Regime · Reg (EU) 2022/858 (2023)	PARTIAL Pilot regime; reform / extension pending	— Venue layer, not an issuer structure	— Venue layer, not an issuer structure

Capability axes.

What the regime can actually deliver: a public chain, a depth of precedent, a listing or bearer form, a settlement money leg, and someone else paying for it.

JURISDICTION & LEGAL BASIS	PUBLIC CHAIN proven by deals	PRECEDENTS depth of issuance	SECONDARY / FORM listing, bearer	CASH LEG settlement money	INCENTIVES who pays
01 France [A]	STRONG SG-FORGE ×4 + EIB on Ethereum mainnet	PARTIAL Several deals, but small private placements	LIMITED [1] Registered + unlisted only in base regime	PARTIAL BdF CBDC experimental; EURCV as workaround	NONE No issuance grant scheme
02 Switzerland [A]	PARTIAL Allowed in law; flagship deals on private SDX	STRONG UBS, World Bank — benchmark size	STRONG SDX listing, legal + proven; no form limit	STRONG SNB wholesale CBDC live on SDX	NONE No issuance grant scheme
03 Luxembourg [B]	LIMITED Allowed; deals on private ledgers	PARTIAL EIB, treasury certificates, LuxSE listings	STRONG LuxSE listing route; CAK / control-agent options	PARTIAL EIB via CBL + BdF DL3S; progressing	NONE No issuance grant scheme
04 Hong Kong [C]	LIMITED Private ledger (CMU / GS DAP / Orion)	STRONG 3 gov issuances incl. largest globally	PARTIAL Benchmark multi-currency done; private rail	STRONG [2] Tokenised CBDC in primary settlement (2025)	STRONG DBGS: 50% of costs, ≤ HK$2.5m
05 Germany [B]	PARTIAL [3] Proven on Polygon, then migrated to permissioned	STRONG KfW / Siemens programme, repeated	STRONG Bearer-native; listable via CSD register	PARTIAL Trigger bridge to RTGS; Pontes pending	NONE No issuance grant scheme
06 United Kingdom [C]	NONE Pilot runs on a private ledger	NONE DIGIT not yet issued (Q1 2027)	PARTIAL Full lifecycle incl. gilts; secondary developing	PARTIAL On-chain settlement in test	NONE [4] Regulatory relief, not a cash grant
07 Singapore [C]	PARTIAL Several Guardian pilots on public chains	PARTIAL OCBC USCP + MAS bill pilots (2025)	PARTIAL Pilot venues; secondary developing	PARTIAL Tokenised deposits; CBDC trials	STRONG [5] G-ADBGS grant scheme, valid to 2029
08 UAE [C]	LIMITED Allowed; FAB deal on a private ledger	PARTIAL One FAB deal (2025)	PARTIAL ADX listing done; private rail	NONE Fiat off-chain settlement	NONE No issuance grant scheme
09 United States [B]	PARTIAL Public Ethereum — funds, not bonds	NONE [6] Deep for funds; essentially none for bonds	PARTIAL ATS for secondary; funds / notes	PARTIAL USDC / tokenised cash; no wholesale CBDC	NONE No issuance grant scheme
— EU-wide [layer]	— Trading / settlement layer, chain-agnostic	LIMITED Modest uptake so far	STRONG This is the trading + settlement layer	PARTIAL CeBM / commercial money / EMT	NONE No issuance grant scheme

Notes on flagged cells.

Six cells carry a qualification that the rating alone would hide.

1 **France · secondary / form.** The national DEEP is registered + unlisted only. Bearer and listed DLT securities are permitted since 2023, but through the EU DLT Pilot Regime, not the base regime.

2 **Hong Kong · cash leg.** The 2025 issuance was the first anywhere to integrate tokenised central bank money (e-HKD, e-CNY) into primary settlement.

3 **Germany · public chain.** Proven on public Polygon, then the flagship programme migrated to permissioned RL1 / SWIAT.

4 **UK · incentives.** No cash grant; the Digital Securities Sandbox offers regulatory facilitation instead.

5 **Singapore · incentives.** Global-Asia Digital Bond Grant Scheme, open to issuers with an Asian nexus, valid to 2029.

6 **US · precedents.** Deep for tokenised funds on public Ethereum; essentially none for bonds specifically.

PART TWO · REGIME DETAIL

II How each jurisdiction regulates a tokenized bond.

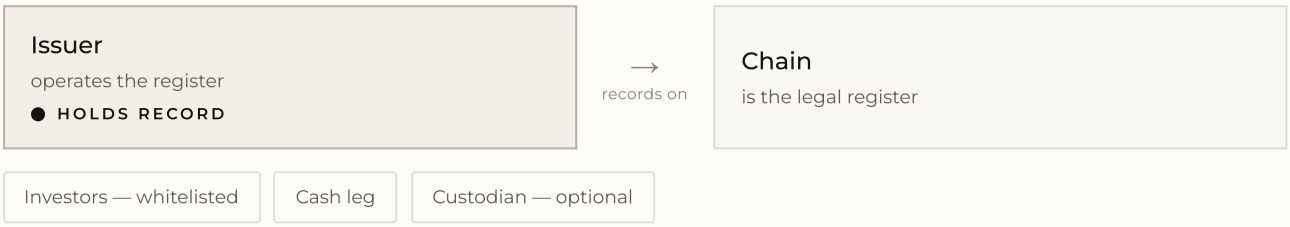
The three infrastructure archetypes show how the parties fit together — each jurisdiction maps to one. Then, per jurisdiction: the issuer's obligations, the documents and infrastructure it needs, the network requirements, cost and subsidies, and the deals that prove it.

Infrastructure archetypes.

Every regime arranges the same building blocks — issuer, record-keeper, chain, custody, cash leg, investors — in one of three ways. The difference is who holds the legal record.

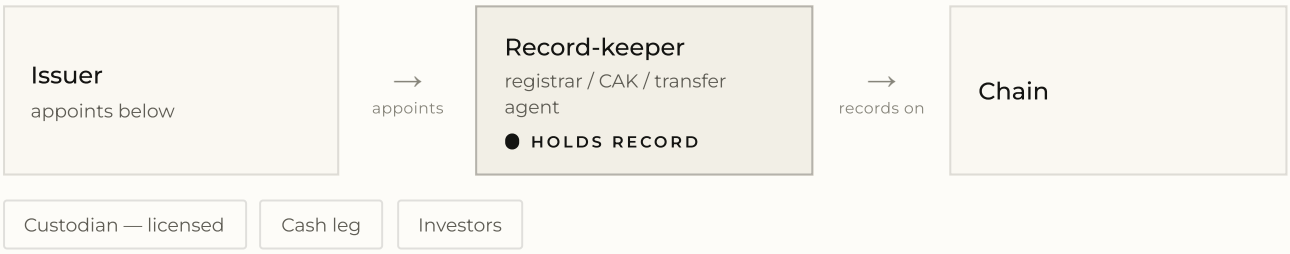
● THE NODE THAT HOLDS THE LEGAL RECORD

A The entry is the record



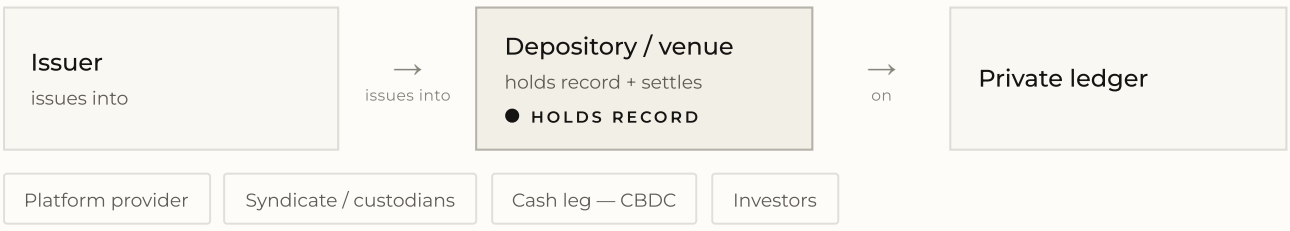
No interposed licence; the issuer keeps the register directly. **France, Switzerland.**

B A licensed record-keeper is interposed



A regulated party answers for the register. **Germany, Luxembourg, US.**

C A regulated depository holds the record



A CSD-like venue operates the ledger and answers for finality. **Hong Kong, UK, UAE, Singapore.**

01 · MODEL B — A LICENSED RECORD-KEEPER IS INTERPOSED

Germany EWPG CRYPTO SECURITIES

REGULATOR	BASIS	PUBLIC CHAIN
BaFin	eWpG §16 (2021) · KWG	proven, then migrated

REGULATION & ISSUANCE

ISSUER MUST

1. Classify the bond as a MiFID II instrument
2. Choose the crypto-security route (eWpG §16), not the central register
3. Appoint a BaFin-licensed crypto securities registrar
4. Appoint a BaFin-licensed crypto custodian
5. Deposit the terms and effect the register entry — which creates the security

DOCUMENTS

- Terms & conditions (Emissionsbedingungen)
- Prospectus, or private-placement exemption memo
- Registrar agreement
- Custody agreement
- ISIN (WM Datenservice)

INFRASTRUCTURE

- **Crypto securities registrar — holds the record**
- Crypto custodian (keys / tokens)
- DLT register (public or permissioned)
- Paying agent
- Cash-leg provider (Bundesbank Trigger → Pontes)

ALSO AML / KYC under the GwG; DORA operational-resilience obligations on ICT providers; the registrar must hold ~€125–150k minimum capital, fit-and-proper management, and an IT-security and compliance organisation. Scope covers dematerialised bearer bonds and, since the Future Financing Act, registered shares.

THE NETWORK

Technology-neutral: the register may be a centralised or a decentralised “tamper-proof system of record,” which includes a public permissionless chain. Accountability sits with the licensed registrar, not the chain, which is what lets a public chain qualify. KfW issued on public Polygon; the flagship programme has since migrated to permissioned RL1 / SWIAT for confidentiality, deterministic finality, and a known validator set.

COST, TIME & INCENTIVES

The heaviest party structure in Europe: a licensed registrar and a licensed custodian must both be engaged, adding cost and coordination relative to France or Switzerland. The framework is mature, so execution is well-trodden; timing turns on onboarding the registrar and custodian. No state subsidy.

PRECEDENTS

KfW issued a €100m crypto security on public Polygon (2024, Cashlink as registrar), a second testing central-bank-money settlement, and a third migrating to SWIAT / RL1 with DekaBank. Siemens ran a crypto-securities programme (2023–24). Deep and repeated.

02 · MODEL A — THE ENTRY IS THE RECORD

France DEEP UNDER PACTE

REGULATOR	BASIS	PUBLIC CHAIN
AMF · ACPR	Ord. 2017-1674 · L211-3 CMF	proven (Ethereum)

REGULATION & ISSUANCE

ISSUER MUST

1. Confirm a MiFID II instrument, unlisted and not CSD-admitted
2. Issue in registered form (nominatif) — bearer unavailable in base DEEP
3. Operate the DEEP itself, or appoint an agent
4. Meet the decree's minimum standards for the DEEP
5. Record the issuance — the registration constitutes the security

DOCUMENTS

- Terms & conditions
- Prospectus, or private-placement exemption
- DEEP operating rules / minimum-standards evidence
- ISIN
- Investor KYC / whitelist records

INFRASTRUCTURE

- **Issuer or its agent as DEEP operator — holds the record**
- The DEEP / public chain
- Account-keeper-custodian — if a third party holds securities
- Arranger / dealer (MiFID II firm)
- Cash-leg provider (euros / BdF CBDC / EURCV)

ALSO AML / KYC with eligibility enforced at the token layer (ERC-3643-style whitelisting); AMF and ACPR oversight; no bespoke registrar licence. For a listing, secondary market, or bearer form, the deal moves to the EU DLT Pilot Regime.

THE NETWORK

The regime is explicitly facilitative: the DEEP's operation does not require a regulated entity, and public permissionless chains qualify. SG-FORGE registered bonds directly on public Ethereum, with investor eligibility enforced at the token layer so the chain stays permissionless.

COST, TIME & INCENTIVES

Among the cheapest and fastest routes — no bespoke registrar to appoint or pay, a private placement avoids a prospectus, and public-chain infrastructure is proven. No state subsidy. The cash leg still relies on off-chain euros, the experimental Banque de France CBDC (DL3S), or SG-FORGE's EURCV stablecoin.

PRECEDENTS

Société Générale / SG-FORGE issued four bonds on public Ethereum mainnet — a €100m covered bond (2019), a €40m issue settled in Banque de France CBDC (2020), and a €10m green bond (2023) — and the EIB's inaugural digital bond (2021) used the French framework. The deepest public-chain bond precedent anywhere, though these are sub-benchmark private placements.

03 · MODEL B — A LICENSED RECORD-KEEPER IS INTERPOSED

Luxembourg BLOCKCHAIN LAWS I–IV

REGULATOR CSSF | BASIS Law of 2013 · Blockchain IV (2024) | PUBLIC CHAIN allowed; private in practice

REGULATION & ISSUANCE

ISSUER MUST

1. Classify as a dematerialised security (Law of 2013)
2. Choose the CAK model or the control-agent model
3. Appoint the regulated record-keeper
4. Notify the record-keeper's role to the CSSF
5. Issue into the DLT issuance account; arrange listing if desired

DOCUMENTS

- Terms & conditions / issuance deed
- Prospectus, or exemption
- Record-keeper appointment + CSSF notification
- LuxSE listing documents — if listed
- ISIN

INFRASTRUCTURE

- **Central account keeper or control agent — holds the record**
- Custodian(s) / account keepers
- The DLT ledger
- Paying agent
- LuxSE (if listed) · cash-leg provider

ALSO AML / KYC; DORA; CSSF supervision; the record-keeper must be an EU credit institution, investment firm, or settlement organisation.

THE NETWORK

Technology-neutral; public chains are permitted, but issuances to date run on private ledgers, such as HSBC Orion for EIB deals. Luxembourg's strength is as a domicile and listing hub rather than a public-chain venue.

COST, TIME & INCENTIVES

Fast to market because the service-provider ecosystem is turnkey, but Luxembourg service costs are premium and a regulated record-keeper is mandatory. No state subsidy.

PRECEDENTS

Multiple EIB digital bonds registered and settled through DLT; the Luxembourg Stock Exchange has admitted DLT and digital bonds to its Securities Official List; the first Luxembourg digital treasury certificates followed in 2025.

04 · MODEL A — THE ENTRY IS THE RECORD

Switzerland DLT ACT, ART. 973D CO

REGULATOR	BASIS	PUBLIC CHAIN
FINMA	DLT Act · CO Art. 973d (2021)	allowed; deals on SDX

REGULATION & ISSUANCE

ISSUER MUST

1. Structure the instrument as a ledger-based security (Art. 973d)
2. Conclude the registration agreement — the constitutive link
3. Ensure the ledger meets Art. 973d(2)
4. Effect the entry, which creates the security
5. For a listing, admit to SDX

DOCUMENTS

- Registration agreement
- Terms & conditions
- Prospectus (if public), or exemption
- Ledger / register documentation
- ISIN

INFRASTRUCTURE

- **Issuer as register operator — holds the record (SDX if listed)**
- The securities ledger (public chain or SDX)
- Investor custodian — optional; direct holding possible
- Paying agent / lead manager
- Cash-leg provider (CHF / SNB wCBDC)

ALSO AML / KYC; FINMA oversight; a “DLT trading facility” licence under FinMIA covers trading venues; non-EU, so EU distribution is handled separately.

THE NETWORK

Technology-neutral, with no France-style registered-only restriction. Public chains are permitted in law; SDX is a permissioned ledger, and flagship deals run there for the regulated venue and central-bank-money settlement.

COST, TIME & INCENTIVES

The leanest legal mechanism for an unlisted issue — a contract creates the security. Listing on SDX adds cost. Switzerland is non-EU, so distribution into the EU needs separate handling. No state subsidy.

PRECEDENTS

A UBS digital bond (2022) and the World Bank’s CHF 200m digital bond (2024), settled in Swiss National Bank wholesale CBDC, both on SDX. Benchmark size, but on a private ledger.

05 · MODEL C — A REGULATED DEPOSITORY HOLDS THE RECORD

Hong Kong EXISTING LAW + SUBSIDIES

REGULATOR HKMA · SFC | BASIS existing HK law + DBGS | PUBLIC CHAIN private ledger

REGULATION & ISSUANCE

ISSUER MUST

1. Structure under existing HK law
2. Engage a tokenisation platform (GS DAP / HSBC Orion)
3. Use the HKMA CMU for clearing and settlement
4. Assemble the syndicate and custodians
5. Apply for the DBGS within 3 months — for the subsidy

DOCUMENTS

- Terms & conditions / offering documents
- Prospectus, or exemption
- Platform / CMU onboarding agreements
- DBGS application
- ISIN

INFRASTRUCTURE

- **HKMA CMU — holds the definitive record + settles**
- Tokenisation platform (GS DAP / HSBC Orion)
- The private blockchain network
- Syndicate banks / investor custodians
- Cash-leg (HKD / e-HKD / e-CNY)

ALSO HKMA and SFC oversight; statutory settlement finality under HK law; for the full DBGS grant, issuance in HK with HK-based lead arrangers and platform.

THE NETWORK

Public chains are allowed, but deals so far run on private ledgers operated through the CMU. Confidentiality, finality, and integrated settlement drive the private-ledger choice.

COST, TIME & INCENTIVES

The Digital Bond Grant Scheme (DBGS, from November 2024) covers 50% of eligible expenses — up to HK$1.25m for a half grant, or HK$2.5m for a full grant meeting additional requirements — for up to two issuances per issuer, and it is open to foreign issuers. This is the only jurisdiction that directly subsidises issuance; green issuances can also draw on the Green and Sustainable Finance Grant Scheme.

PRECEDENTS

HKSAR government tokenised green bonds in 2023 (HK$800m, the first government tokenised bond globally), 2024 (~HK$6bn, multi-currency), and 2025 (~HK$10bn / US$1.3bn — the largest digital bond globally, and the first anywhere to integrate tokenised central bank money, e-HKD and e-CNY, into primary settlement).

06 · MODEL C — A REGULATED DEPOSITORY HOLDS THE RECORD

United Kingdom DIGITAL SECURITIES SANDBOX

REGULATOR Bank of England + FCA | BASIS DSS (2024) | PUBLIC CHAIN private ledger

REGULATION & ISSUANCE

ISSUER MUST

1. Enter the DSS regime (BoE + FCA)
2. Use a DSS-approved Digital Securities Depository
3. Confirm the DSD has reached Gate 2 (live)
4. Arrange an investor-side DSD link if needed
5. Stay within scope caps (corporate ≤ £900m)

DOCUMENTS

- Terms & conditions
- Prospectus / admission documents
- DSS participation / DSD agreements
- ISIN
- Investor onboarding records

INFRASTRUCTURE

- **Digital Securities Depository — holds the record (HSBC Orion)**
- Investor DSD (e.g. LSEG) for the link
- The private ledger
- Paying agent / lead manager
- Cash-leg (central bank money; on-chain in test)

ALSO BoE and FCA joint oversight; the sandbox is temporary and gate-based; the incentive is regulatory facilitation, not a cash grant.

THE NETWORK

The pilot runs on a private ledger; public chains are not part of the current design.

COST, TIME & INCENTIVES

Sandbox admission and a small set of approved operators add process and time, and the DSS is a temporary regime rather than permanent market infrastructure. No cash grant — the incentive is regulatory facilitation.

PRECEDENTS

HSBC Orion became the first live DSD (Gate 2, July 2026). DIGIT, the UK's first sovereign digital gilt, is expected in Q1 2027, with HSBC as platform provider and an LSEG connectivity link. HSBC Orion has enabled over US$5bn of digital bond issuance globally.

07 · MODEL B — A LICENSED RECORD-KEEPER IS INTERPOSED

United States EXISTING SECURITIES LAW

REGULATOR	BASIS	PUBLIC CHAIN
SEC · FINRA	existing securities law	proven — funds, not bonds

REGULATION & ISSUANCE

ISSUER MUST

1. Determine the security type and exemption (Reg D / Reg S)
2. Appoint an SEC-registered transfer agent
3. Appoint a qualified custodian
4. Engage a broker-dealer; for a bond, an indenture trustee
5. Enforce whitelisted transfers + off-chain identity linkage

DOCUMENTS

- Offering memorandum / PPM
- Reg D / Reg S filings (Form D)
- Transfer-agent + custody agreements
- Indenture — for a bond
- Investor accreditation / KYC records

INFRASTRUCTURE

- **SEC-registered transfer agent — holds the record (public Ethereum)**
- Qualified custodian
- Public chain (ERC-20 + permissioned token logic)
- Broker-dealer / ATS (for secondary)
- Cash-leg (USD / USDC)

ALSO SEC and FINRA oversight; ATS registration for a secondary market; existing securities law applies unchanged — no bespoke DLT regime.

THE NETWORK

The record is kept on public Ethereum, and other public chains, with a permissioned framework at the token layer. This is genuine public-chain issuance of regulated securities — for money-market and treasury funds and notes, not for bonds.

COST, TIME & INCENTIVES

Heavy structuring and expensive securities counsel, with more parties than the European light regimes. No state subsidy.

PRECEDENTS

Deep for tokenised money-market and treasury funds on public Ethereum — BlackRock's BUIDL with Securitize, BSTBL with BNY Mellon as transfer agent, Franklin Templeton's BENJI, Ondo — with secondary trading via registered alternative trading systems. Essentially no tokenised corporate or government bonds.

08 · MODEL C — A REGULATED DEPOSITORY HOLDS THE RECORD

UAE SCA / DFSA / FSRA

REGULATOR SCA · DFSA · FSRA | BASIS frameworks in force | PUBLIC CHAIN private ledger

REGULATION & ISSUANCE

ISSUER MUST

1. Choose onshore (SCA / ADX) or a free zone (DFSA / FSRA)
2. Engage HSBC Orion (via CMU) as issuance / tokenisation platform
3. List on ADX — onshore route
4. Arrange ICSD links (Euroclear / Clearstream / CMU)
5. Appoint a global coordinator / bookrunner

DOCUMENTS

- Terms & conditions / offering documents
- ADX listing / admission documents
- Platform onboarding agreements
- ISIN
- Investor KYC records

INFRASTRUCTURE

- **HSBC Orion (via CMU) — holds the record + platform**
- ADX — listing venue
- The private ledger
- ICSD links (Euroclear / Clearstream / CMU)
- Cash-leg (fiat)

ALSO SCA onshore or DFSA / FSRA free-zone rules; standard bond disclosure; the bond-specific path is new (one deal to date).

THE NETWORK

Public chains are allowed, but the flagship deal ran on a private ledger (HSBC Orion).

COST, TIME & INCENTIVES

Business-friendly and well-resourced, but the bond-specific path is new, with a single deal to date. No dedicated issuance subsidy identified.

PRECEDENTS

First Abu Dhabi Bank issued US$100m three-year floating-rate notes (July 2025) — the first digitally-native bond in MENA — listed on ADX via HSBC Orion, with HSBC as sole global coordinator.

09 · MODEL C — A REGULATED DEPOSITORY HOLDS THE RECORD

Singapore TECH-NEUTRAL SFA; MAS PILOTS

REGULATOR MAS | BASIS SFA · Guardian / GL1 | PUBLIC CHAIN some pilots public

REGULATION & ISSUANCE

ISSUER MUST

1. Structure under the tech-neutral SFA
2. Choose a platform / GL1 or a Project Guardian venue
3. For the grant, joint-lead with ≥2 MAS-licensed entities
4. Assemble standard roles (arranger, custodian, paying agent)
5. Apply for G-ADBGS — for the subsidy

DOCUMENTS

- Terms & conditions / offering documents
- Prospectus, or exemption
- Platform onboarding agreements
- G-ADBGS application
- ISIN

INFRASTRUCTURE

- **Ledger operator / platform — holds the record**
- The chain (public in pilots; GL1 permissioned)
- Arranger / custodian
- Paying agent
- Cash-leg (SGD / tokenised deposits / CBDC trials)

ALSO MAS oversight; the SFA applies as to any security; G-ADBGS eligibility — Asian nexus, minimum S$100m (S$200m+ digitally native), up to two issuances.

THE NETWORK

Mixed: several Project Guardian pilots use public chains (Ethereum, Polygon, Avalanche), while GL1 is a permissioned shared ledger built for institutional scale.

COST, TIME & INCENTIVES

The Global-Asia Digital Bond Grant Scheme (G-ADBGS, from January 2025, valid to 2029) subsidises qualifying digital bond issuances — open to issuers with an Asian nexus, minimum S$100m — for up to two issuances. Operationally the market is still pilot-stage.

PRECEDENTS

Live Project Guardian use cases across fixed income and funds; OCBC's tokenised US-dollar commercial paper on J.P. Morgan's Kinexys (2025, settled within minutes); and MAS tokenised-bill pilots settled in CBDC (2025).

10 · THE EU TRADING & SETTLEMENT LAYER

EU-wide DLT PILOT REGIME

REGULATOR NCAs + ESMA opinion | BASIS Reg (EU) 2022/858 (2023) | PUBLIC CHAIN chain-agnostic layer

REGULATION & ISSUANCE

TO REACH IT, USE

1. A licensed DLT market infrastructure
2. A DLT-MTF (trading), DLT-SS (settlement), or DLT-TSS (both)
3. National authorisation, with a non-binding ESMA opinion
4. Compliance within the threshold caps

DOCUMENTS

- Operator authorisation application
- Admission documents for the instrument
- Exemption justifications (MiFID II / CSDR)
- ISIN

INFRASTRUCTURE

- **DLT-MTF / SS / TSS operator — holds the record + settles**
- The DLT (chain-agnostic)
- Cash-leg (CeBM / commercial money / EMT)
- Sits above the national register model

ALSO this is a venue authorisation, not a national register regime; it licenses operators, not issuers.

THE NETWORK

A chain-agnostic trading and settlement layer that sits above the national register models rather than replacing them.

COST, TIME & INCENTIVES

Thresholds cap bonds at under €1bn per issuance, with a €6bn aggregate per infrastructure and an exit strategy triggered at €9bn. Uptake has been modest, and a reform to widen eligible operators and scope is pending, expected around end-2027. No subsidy.

PRECEDENTS

A handful of authorised DLT market infrastructures — 21X as a DLT-TSS, CSD Prague as a DLT-SS, 360X as a DLT-MTF. This is where EU secondary trading and settlement at scale is licensed.

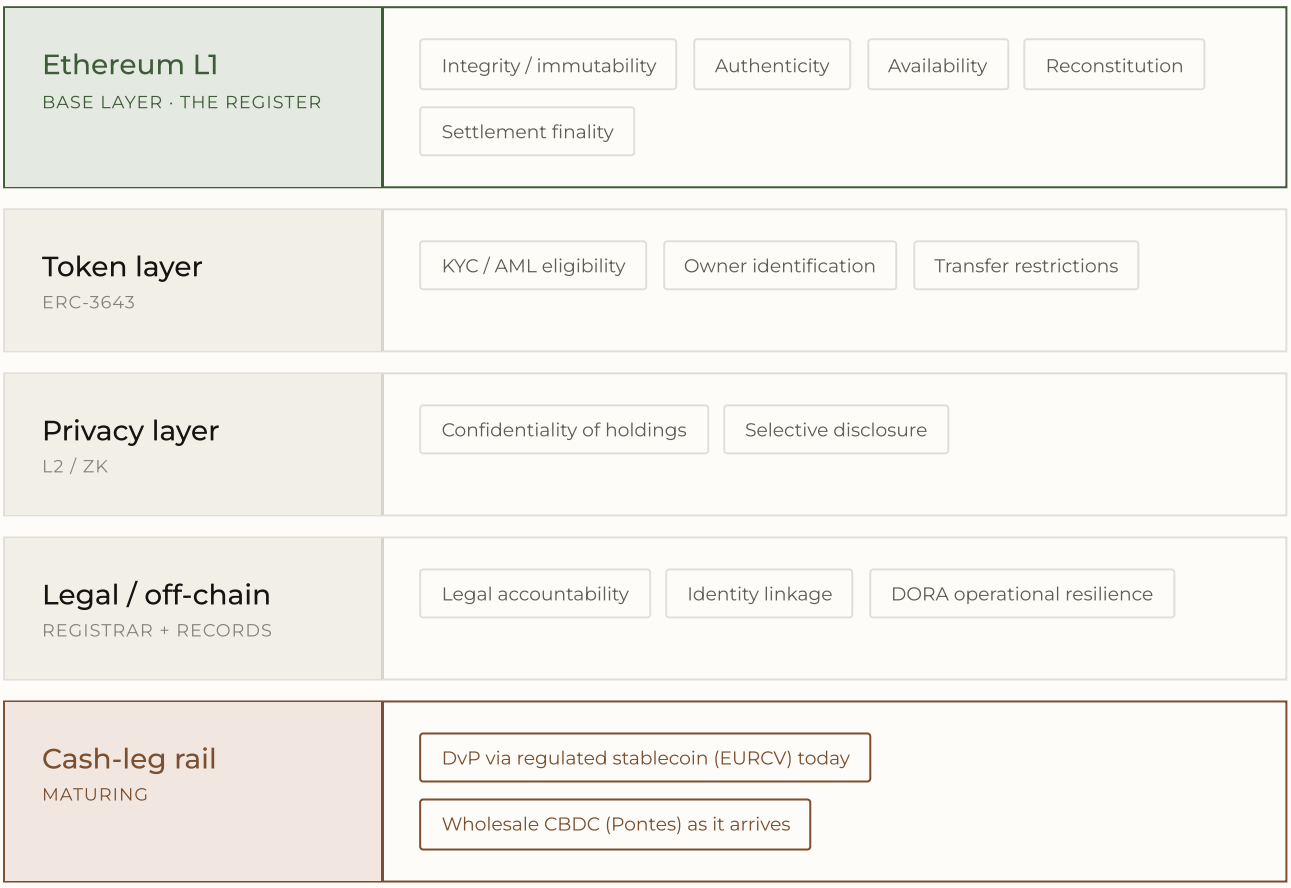
PART THREE · ETHEREUM

III How the requirements are met on public Ethereum.

The technology-neutral regimes — Germany's eWpG, France's DEEP, Switzerland's DLT Act, Luxembourg's Blockchain Laws — already permit a public permissionless chain to carry the legal register. Every requirement above is satisfied somewhere in the stack: the base chain holds the register, the layers above it carry compliance, privacy, and accountability.

The compliance stack.

Read from the foundation up. The base chain qualifies as the register; the token layer enforces eligibility; a privacy layer restores confidentiality; a regulated party answers off-chain. The cash leg is the one piece still maturing.



The register requirement is met at L1; compliance and confidentiality sit in the layers above; accountability sits with a regulated party off-chain. The cash leg is the remaining piece, and the same layered logic closes it — a regulated stablecoin now, wholesale central bank money as it arrives.

Requirement → how it is met.

Eleven requirements, and the layer of the stack that satisfies each one.

REQUIREMENT	SATISFIED AT	HOW
Integrity / immutability of the register	L1	Cryptographic hashing and consensus replication; entries cannot change without network agreement.
Authenticity of entries	L1 · TOKEN	Every transfer is cryptographically signed by the holder's key.
Availability and continuity	L1	Roughly ten years with no full-network outage; globally distributed nodes.
Reconstitution of the register	L1	Full state replicated across nodes; the register rebuilds from the chain.
Identification of owners	TOKEN · OFF-CHAIN	ERC-3643 permissioned tokens; wallets linked to KYC'd identities off-chain, as US transfer agents do.
KYC / AML eligibility	TOKEN	Whitelisted addresses; transfer rules enforced in the contract while the chain stays permissionless.
Confidentiality of holdings	PRIVACY L2	ZK validium (Prividium), Aztec, selective disclosure — proofs without exposing the underlying data.
Settlement finality	L1 · L2	~12–13 min economic finality on L1; sub-second soft finality on L2; sufficient for bond workflows.
Legal accountability	OFF-CHAIN	The licensed registrar or issuer answers for the register; the chain is the medium, not the accountable party.
Operational resilience (DORA)	OFF-CHAIN	ICT-risk obligations sit on the registrar and operator, not the base protocol.
Delivery-versus-payment — cash leg	CASH-LEG	Regulated stablecoin (EURCV) for on-chain DvP today; wholesale CBDC (Pontes) as it arrives.

The verdict.

Every register requirement is met at L1; eligibility and identification at the token layer; confidentiality at a privacy L2; accountability off-chain with the registrar. SG-FORGE proved this for bonds on public Ethereum in France, and US transfer agents run regulated fund shares on public Ethereum today. The remaining gap is the cash leg, and the same layered logic closes it — a regulated stablecoin now, wholesale central bank money as it arrives.

Public Ethereum satisfies the regulatory requirements for a tokenized bond: the register at L1, compliance at the token layer, confidentiality at a privacy L2, accountability off-chain. The cash leg is the one piece still maturing, and it is closing.

SCALE

Strong / Partial / Limited / None. Friction axes (clarity, ease, cost) run favourable → heavy; capability axes (public chain, precedents, cash leg, incentives) run present → absent. Each matrix cell and each regime brief states its basis.

SCOPE

Institutional / privately-placed digital bonds. Not legal advice — a real mandate turns on instrument form, investor base, and distribution.

COMPANIONS

The applied schema (DE / FR stack) · the three record-keeper models · deal structures by jurisdiction.

This report is provided for information only. It does not constitute investment, legal, tax or accounting advice, and it is not an offer or solicitation. Data is stated as at August 2026 and has not been independently verified. A real mandate turns on instrument form, investor base, and distribution.



ABOUT THE AUTHOR

Adam Berker

I'm a Crypto Regulatory Counsel with six years across regulated crypto and DeFi. My work sits at the translation layer between regulators and builders: fitting securities, custody, and settlement structures to a jurisdiction, and putting on-chain mechanics in terms a regulator can supervise.

Questions on any of this — find me here.

EMAIL adam@berker.legal

LINKEDIN /in/adamberker

SELECTED TRACK RECORD

- → Obtained and managed 10+ crypto and payment licenses
- → Built regulated crypto activity compliant with MiCA, DORA, UK FinProm, and US MTL, and expanded it across the US, Canada, and LatAm
- → Built cross-jurisdictional structures combining US, EU, and UK entities with offshore foundations and associations
- → Structured token raises via SAFE, SAFT, and token warrants, and closed a $40M Series A/B
- → Concluded 50+ partnerships and vendor contracts with tier-1 payment networks, exchanges, and protocols

HOW I CAN HELP

- → **Licensing** — which licenses a structure actually needs, and obtaining them
- → **Regulatory analysis** — SEC, FCA, MiCA, DORA, US MTL: what applies and what does not
- → **Regime fit** — matching an instrument to the right jurisdiction, and flagging where it leaves the base framework
- → **Issuance structuring** — a tokenized bond or fund end to end: register, custody, paying agent, cash leg
- → **Public-chain compliance** — showing how a structure meets a regulator's requirements, and designing the layers that close the gaps
- → **Corporate structuring** — cross-jurisdictional setups with US entities, offshore foundations, and associations
- → **Token launches & fundraising** — SAFE, SAFT, token warrants, and a structure that holds up under scrutiny
- → **Commercial deals** — partnerships, integrations, and vendor contracts

ON-CHAIN BOND ISSUANCE

A regulatory schema for institutional digital bonds: ten jurisdictions scored on eight axes, the regime behind every rating, and how each requirement is satisfied on public Ethereum.

Adam Berker